3.1 Factoring $ax^2 + bx + c$

Name _____ Period ____ Date _____

Algebra II 3.1 (30-minute plan) · Student sheet

I can…

- factor out the GCF and factor $ax^2 + bx + c$.
- solve quadratic equations by factoring.
- find x-intercepts and solve real-life problems by factoring.

Vocabulary · fill in the missing word

Greatest common factor (GCF) — the greatest factor _____ by every term

Factoring — writing an _____ as a product

Zero-Product Property — if ab = _____, then a = 0 or b = 0

Zero of a function — an _____ where f(x) = 0; an x-intercept of the graph

Part 1 · Notes and examples (fill in each note, then work the example)

Factor the GCF: Always factor out the _____ first.

Example 1

Factor out the GCF: $6x^2 - 15x$

Factoring $ax^2 + bx + c$: For $ax^2 + bx + c$, find factors of _____ that add to _____.

Example 2

Factor each expression.

a. $x^2 - 9x + 20$ b. $2x^2 + 5x - 12$

Zero-Product Property: Zero-Product Property: if ab = 0, then _____.

Example 3

Solve $3x^2 - 5x - 2 = 0$.

Part 2 · Partner practice

Partner practice

Factor $4x^2 - 4x - 3$.